Structural BMPs

Purpose

This page documents the structural BMP parameter files referenced by HRUs and landscape units: `tiledrain.str`, `septic.str`, `filterstrip.str`, `grassedww.str`, and `bmpuser.str`. File names for the first five are set in `type input_structural` in `src/input_file_module.f90`. The page also covers two closely related files declared elsewhere: `septic.sep` (effluent quality, named in `type input_parameter_databases`), `urban.urb` (urban land-cover parameters, same type), and `satbuffer.str` (saturated buffer, file name hard-coded in the reader).

Each file is optional. If the slot in `file.cio` is `null` or the file is missing, the reader allocates a zero-length array and skips on. Every reader skips two header lines, then reads one record per parameter set.

Records in `tiledrain.str`, `filterstrip.str`, `grassedww.str`, and `bmpuser.str` are referenced by name from `hru-data.hru` (columns `tiledrain`, `vfsfield`, `grassww`, `bmpuser`). Records in `septic.str` are referenced from the management schedule when a septic operation is scheduled.

Source

- File names: src/input_file_module.f90, `type input_structural`, `type input_parameter_databases`
- Readers:
  - src/sdr_read.f90 (`tiledrain.str`)
  - src/sep_read.f90 (`septic.str`)
  - src/septic_parm_read.f90 (`septic.sep`)
  - src/scen_read_filtstrip.f90 (`filterstrip.str`)
  - src/scen_read_grwway.f90 (`grassedww.str`)
  - src/scen_read_bmpuser.f90 (`bmpuser.str`)
  - src/sat_buff_read.f90 (`satbuffer.str`)
  - src/urban_parm_read.f90 (`urban.urb`)

tiledrain.str

Subsurface drain parameter sets. Read into `sdr` (`type subsurface_drainage_parameters` in `src/hru_module.f90`). Referenced by the `tiledrain` column of `hru-data.hru`.

#	Column	Type	Units	Default	Description
1	name	char(40)	none	“null”	parameter set name
2	depth	real	mm	0.0	depth of drain tube from soil surface
3	time	real	hours	0.0	time to drain soil to field capacity
4	lag	real	hours	0.0	drain tile lag time
5	radius	real	mm	0.0	effective radius of drains
6	dist	real	mm	0.0	distance between two drain tubes or tiles
7	drain_co	real	mm/day	0.0	drainage coefficient
8	pumpcap	real	mm/hr	0.0	pump capacity
9	latksat	real	none	0.0	multiplication factor for lateral saturated hydraulic conductivity

Example from `refdata/Osu_1hru/tiledrain.str`:

```
tiledrain.str: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                          dp          t_fc           lag           rad          dist         drain
mw24_1000           1000.00000      24.00000      96.00000     100.00000      30.00000      10.00000
```

Editor column names (`dp`, `t_fc`, `rad`, `drain`, `pump`, `lat_ksat`) differ from source field names. The reader uses position only.

septic.str

Per-HRU septic system parameter sets (geometry, biozone properties, decay coefficients). Read into `sep` (`type septic_system` in `src/septic_data_module.f90`).

#	Column	Type	Units	Description
1	name	char(13)	none	parameter set name
2	typ	int	none	septic system type
3	yr	int	year	year the septic system became operational
4	opt	int	none	operation flag (1 = active, 2 = failing, 0 = not operated)
5	cap	real	none	number of permanent residents
6	area	real	m^2	average drainfield area per septic system
7	tfail	int	days	time until failing system gets fixed
8	z	real	mm	depth to top of biozone layer
9	thk	real	mm	thickness of biozone layer
10	strm_dist	real	km	distance to stream
11	density	real	per km^2	number of septic systems per square kilometre
12	bd	real	kg/m^3	density of biomass
13	bod_dc	real	m^3/day	BOD decay rate coefficient
14	bod_conv	real	none	mass conversion factor between bacterial growth and BOD degraded
15	fc1	real	none	linear coefficient for biozone field capacity
16	fc2	real	none	exponential coefficient for biozone field capacity
17	fecal	real	m^3/day	fecal coliform decay rate coefficient
18	plq	real	none	plaque conversion factor from TDS
19	mrt	real	none	mortality rate coefficient
20	rsp	real	none	respiration rate coefficient
21	slg1	real	none	slough-off calibration parameter
22	slg2	real	none	slough-off calibration parameter
23	nitr	real	none	nitrification rate coefficient
24	denitr	real	none	denitrification rate coefficient
25	pdistrb	real	L/kg	linear P sorption distribution coefficient
26	psorpmax	real	mg P/kg soil	maximum P sorption capacity
27	solpslp	real	none	slope of linear effluent soluble P equation
28	solpintc	real	none	intercept of linear effluent soluble P equation

Example from `refdata/Osu_1hru/septic.str`:

```
septic.str: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                    typ         yr   operation      residents           area     t_fail        dp_bioz       t
standard                  1          0           0        2.50000      100.00000         70      500.00000      50
failing                   1          0           0        2.50000      100.00000          1      500.00000      50
```

septic.sep

Effluent quality database. One record per effluent type. Records are referenced by index from elsewhere (no name resolution in the reader). Read into `sepdb` (`type septic_db` in `src/septic_data_module.f90`). File name is in `in_parmdb%septic_sep`.

#	Column	Type	Units	Description
1	sepnm	char(20)	none	effluent set name
2	qs	real	m^3/day	flow rate per capita
3	bodconcs	real	mg/L	biological oxygen demand
4	tssconcs	real	mg/L	total suspended solids
5	nh4concs	real	mg/L	NH4-N concentration
6	no3concs	real	mg/L	NO3-N concentration
7	no2concs	real	mg/L	NO2-N concentration
8	orgnconcs	real	mg/L	organic N concentration
9	minps	real	mg/L	mineral P concentration
10	orgps	real	mg/L	organic P concentration
11	fcolis	real	mg/L	fecal coliform concentration

The source comment on `nh4concs` reads "concentration of total phosphorus"; the field is the NH4-N concentration based on column position and downstream use. Example from `refdata/Osu_1hru/septic.sep` (first two records shown):

```
septic.sep: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                      q_rate           bod           tss          nh4_n          no3_n          no2_n
gcon                     0.22700     170.00000      75.00000      42.40000        0.00000        0.00000
gadv                     0.22700      22.00000      14.00000      18.90000        9.60000        0.00000
```

filterstrip.str

Vegetative filter strip parameter sets. Read into `filtstrip_db` (`type filtstrip_operation` in `src/mgt_operations_module.f9`
Referenced by the `vfsfield` column of `hru-data.hru`.

#	Column	Type	Units	Default	Description
1	name	char(40)	none	“”	parameter set name
2	vfsi	int	none	0	on/off flag for vegetative filter strip
3	vfsratio	real	none	0.0	drainage-area to filter-strip-area ratio
4	vfscon	real	none	0.0	fraction of total runoff from the entire field
5	vfsch	real	none	0.0	fraction of flow entering the most concentrated 10% of the VFS that is fully channelized

Example from `refdata/Osu_1hru/filterstrip.str`:

```
filterstrip.str: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                  flag       fld_vfs       con_vfs         cha_q  description
field_border             0       0.10000       0.00300       0.20000  Field_border
high_engineered          0       0.10000       0.00100       0.05000  Highly_engineered_low_channelized
```

grassedww.str

Grassed waterway parameter sets. Read into `grwaterway_db` (`type grwaterway_operation` in `src/mgt_operations_module.f9`
Referenced by the `grassww` column of `hru-data.hru`.

#	Column	Type	Units	Default	Description
1	name	char(40)	none	“”	parameter set name
2	grwat_i	int	none	0	on/off flag for waterway simulation
3	grwat_n	real	none	0.0	Manning’s n for grassed waterway
4	grwat_spcon	real	none	0.0	sediment transport coefficient
5	grwat_d	real	m	0.0	depth of grassed waterway
6	grwat_w	real	m	0.0	width of grassed waterway
7	grwat_l	real	km	0.0	length of grassed waterway
8	grwat_s	real	m/m	0.0	slope of grassed waterway

Example from `refdata/Osu_1hru/grassedww.str`:

```
grassedww.str: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                  flag          mann        sed_co            dp            wd           len
grwway_high              0       0.05000       0.02000       1.00000       4.00000       0.50000       0.1
grwway_med               0       0.05000       0.02000       0.75000       3.00000       0.75000       0.0
grwway_low               0       0.05000       0.02000       0.50000       2.00000       1.00000       0.0
```

bmpuser.str

User-defined upland BMP removal-efficiency sets. Read into `bmpuser_db` (`type bmpuser_operation` in `src/mgt_operations_module.f90`). Referenced by the `bmpuser` column of `hru-data.hru`.

#	Column	Type	Units	Default	Description
1	name	char(40)	none	“”	parameter set name
2	bmp_flag	int	none	0	on/off flag
3	bmp_sed	real	percent	0.0	sediment removal
4	bmp_pp	real	percent	0.0	particulate P removal
5	bmp_sp	real	percent	0.0	soluble P removal
6	bmp_pn	real	percent	0.0	particulate N removal
7	bmp_sn	real	percent	0.0	soluble N removal
8	bmp_bac	real	percent	0.0	bacteria removal

Example from `refdata/Osu_1hru/bmpuser.str`:

```
bmpuser.str: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                  flag       sed_eff      ptlp_eff      solp_eff      ptln_eff      soln_eff      bact
bmpusr1                  1       0.20000       0.20000       0.20000       0.20000       0.20000       0.2
```

Note: a second type `bmpuser_operation1` in the same module exposes a wider parameter set (surface, subsurface, tile flow and constituents). The reader populates `bmpuser_db` (`bmpuser_operation`); the wider type is not read here.

satbuffer.str

Saturated buffer source-receiver pairs. Read into `satbuff_db` (`type saturated_buffer_parameters` in `src/hru_module.f90`). The file name is hard-coded as `"satbuffer.str"` in the reader, not listed in `type input_structural`. The file is loaded only when it exists.

#	Column	Type	Description
1	name	char(40)	parameter set name
2	hru_src	int	source HRU contributing tile inflow
3	frac_src	real	fraction of source HRU contributing to tile flow
4	flocon_dtbl	char(40)	flow-control decision table name (resolved against `dtbl_flo`)
5	hru_rcv	int	receiving (buffer) HRU
6	lyr	int	soil layer for incoming tile flow (0 = surface)

After reading, the reader copies the parameter set into `hru(hru_src)%sb%sb_db` and `hru(hru_rcv)%sb%sb_db`, then resolves `flocon_dtbl` against the loaded flow-control decision tables (`dtbl_flo`) to set `hru(hru_src)%sb%dtbl`.

urban.urb

Urban land-cover parameter database (build-up, wash-off, percent impervious, urban CN). Read into `urbdb` (`type urban_db` in `src/urban_data_module.f90`). File name is in `in_parmdb%urban_urb`. Urban parameter sets are referenced from `landuse.lum` for HRUs flagged as urban.

#	Column	Type	Units	Default	Description
1	urbnm	char(16)	none	“”	urban land-cover name
2	fimp	real	fraction	0.05	fraction of HRU area that is impervious
3	fcimp	real	fraction	0.05	fraction of HRU classified as directly connected impervious
4	curbden	real	km/ha	0.0	curb length density
5	urbcoef	real	1/mm	0.0	wash-off coefficient for removal of constituents from an impervious surface
6	dirtmx	real	kg/curb-km	1000.0	maximum amount of solids allowed to build up on impervious surfaces
7	thalf	real	days	1.0	time for solids on impervious areas to build up to half the maximum
8	tnconc	real	mg N / kg sed	0.0	total nitrogen concentration in suspended-solid load from impervious areas
9	tpconc	real	mg P / kg sed	0.0	total phosphorus concentration in suspended-solid load from impervious areas
10	tno3conc	real	mg NO3-N / kg sed	0.0	NO3-N concentration in suspended-solid load from impervious areas
11	urbcn2	real	none	98.0	moisture condition II curve number for impervious areas

Example from `refdata/Osu_1hru/urban.urb` (first three records shown):

```
urban.urb: written by SWAT+ editor v2.2.0 on 2023-03-22 04:25 for SWAT+ rev.60.5.4
name                  frac_imp   frac_dc_imp      curb_den      urb_wash      dirt_max     t_halfmax     c
urhd                   0.60000       0.44000       0.24000       0.18000     225.00000       0.75000     5
urmd                   0.38000       0.30000       0.24000       0.18000     225.00000       0.75000     5
urml                   0.20000       0.17000       0.24000       0.18000     225.00000       0.75000     4
```

Related files

- file.cio. The `structural` and `hru_parm_db` lines select which of these files are read.
- hru-data.hru. References `tiledrain.str`, `filterstrip.str`, `grassedww.str`, and `bmpuser.str` by name.
- landuse.lum. References `urban.urb` entries for urban HRUs.